HDOC Day-56

Maximum Subarray Sum Using Kadane's Algorithm

Aim

To write a C program to find the maximum sum of any contiguous subarray using Kadane's algorithm, while passing the array to a function using its address.

Problem Statement

- Read the size and elements of a variable-length integer array, arr[].
- Pass the array to max_subarray_sum() using its address.
- Use pointers to pass and access the array.
- Find the maximum contiguous subarray sum using Kadane's algorithm.
- If all elements are negative, display the largest (least negative) element.
- Ensure that the selected subarray contains at least one element.

Algorithm

1. Read the size and elements of the array.

2. Pass the array address and size to max_subarray_sum().

3. Initialize current_sum and max_sum with the first element.

4. For each remaining element, either start a new subarray from that element or extend the current subarray.

5. Update max_sum whenever current_sum is greater.

6. Display max_sum.

C Program

```c
#include <stdio.h>

void max_subarray_sum(int *arr, int size)
{
    int current_sum = *arr;
    int max_sum = *arr;
    int i;

    for (i = 1; i < size; i++)
    {
        if (current_sum + *(arr + i) > *(arr + i))
            current_sum = current_sum + *(arr + i);
        else
            current_sum = *(arr + i);

        if (current_sum > max_sum)
            max_sum = current_sum;
    }

    printf("Maximum subarray sum: %d\n", max_sum);
}

int main()
{
    int size, i;

    printf("Enter size of array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid size\n");
        return 0;
    }

    int arr[size];

    printf("Enter array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", &arr[i]);

    max_subarray_sum(arr, size);
    return 0;
}
```

Test Cases

Test	Input Array	Expected Output
1	-2 1 -3 4 -1 2 1 -5 4	Maximum subarray sum: 6
2	-8 -3 -6 -2 -5 -4	Maximum subarray sum: -2
3	5 -2 3 4	Maximum subarray sum: 10

Sample Outputs

Test Case 1 `Enter size of array: 9\nEnter array elements: -2 1 -3 4 -1 2 1 -5 4\nM`

Test Case 2 `Enter size of array: 6\nEnter array elements: -8 -3 -6 -2 -5 -4\nMaxim`

Test Case 3	`Enter size of array: 4\nEnter array elements: 5 -2 3 4\nMaximum subarr`

Result

The program successfully finds the maximum sum of a contiguous subarray using Kadane's algorithm and correctly handles the case when all elements are negative.